Using Graphkit-Genco Grid Foundation Model to Recover the Aged Admittance Matrix

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November 3, 2026

Abstract

The grid admittance matrix is a critical, confidential input used in almost all grid calculations, including power flow, optimal power flow, and state estimation. However, this matrix changes over time as the grid ages. Accurately tracking changes in line admittances can be challenging, especially when direct measurements are unavailable or coverage is limited. We propose using high-fidelity grid foundation models to estimate the aged admittance matrix. We use the GENCO model and assume that the grid operator has access to its nominal admittance matrix. We first show numerically that, using the nominal admittance matrix, the GENCO-estimated voltage magnitude and phase at buses are not biased and have symmetric error distributions around zero. However, using a fine-tuned GENCO model with the nominal admittance matrix to predict masked voltage amplitudes and phases from data samples generated with an aged admittance matrix yields skewed error probability density functions. We then propose a gradient-based procedure that estimates the aged admittance matrix by optimizing line impedances through a frozen GENCO model. We evaluate the accuracy of the estimated admittance matrix and demonstrate that it reduces the dispersion and systematic shift of voltage magnitude and phase-angle prediction errors.

1 Introduction

Grid Foundation Models (Grid-FM) offer a promising, transformative approach to addressing bottlenecks in the power industry's adoption of Artificial Intelligence. Grid-FMs use a generic representation of power-network operational data and topology, along with a physics-informed model architecture to overcome the limited transferability of developed models to adjacent applications and system configurations. They can be used across a broad range of applications and address limited data at each utility by enabling training of high-fidelity models [3].

This paper uses the Genco (Geometric Neural Corrective Optimization) foundation model, which provides a unified framework for optimal power flow (OPF), power flow (PF), and state estimation (SE) [5]. Genco is trained on diverse synthetic data generated by gridfm-datakit [6]. It uses a designed shared grid representation that could be used by three applications. Genco uses large-scale datasets with millions of PF and OPF scenarios and was evaluated on benchmark datasets.

GENCO deploys a physics-informed neural solver developed to accelerate steady-state transmission network analysis. OPF, PF, and SE are nonlinear and nonconvex and were traditionally solved with iterative numerical methods, each with its own solution approach. However, GENCO addresses these well-known power network problems simultaneously in a single shared model representation, using a common architecture that combines graph transformers, physics-constraint decoding, and iterative corrections guided by power balance residuals across heterogeneous graph neural network (GNN) layers. This design aims to improve computational efficiency while maintaining physical consistency across different network configurations and operating conditions, moving toward grid foundation models.

Training the GENCO model in a network-architecture-agnostic way and under different perturbations makes it robust to unseen networks and conditions. GENCO has strong potential for future use in power network studies. For example, the GENCO model can eliminate the need to install expensive measurement units by providing a relatively accurate estimate of the network state or, as we discuss in this paper, evaluating the consistency between grid operator data and measurements.

Aging in power networks refers to the progressive deterioration of equipment from thermal, electrical, mechanical, and environmental stresses over time, which can reduce reliability and change equipment characteristics. For example, in generators, mechanical and thermal stresses degrade winding insulation, whereas in circuit breakers, repeated operation erodes contact points and reduces switching reliability. Transmission lines also age from exposure to extreme weather and thermal stress, causing conductor corrosion, material deterioration, and drift in electrical impedance. In an aged transmission line, impedance characteristics increase from their nominal values; both resistance and reactance are higher than their nominal values. As a consequence of line aging, losses increase, so some energy is dissipated across the line during transmission.

The aging effects of transmission lines extend beyond power loss; they also create challenges for downstream power system analyses. Tasks such as SE and stability assessment rely on accurate transmission line impedance values. Uncertainty in line impedance can compromise assessments of operating-point feasibility and lead to unreliable evaluations of risks associated with proposed power-dispatch solutions. Continuous measurements and monitoring are also expensive. Therefore, it is paramount to find a cheap, viable method to estimate line impedance drift due to aging.

This study proposes using high-fidelity grid foundation models to first identify potential deviations from the nominal admittance matrix due to aging and subsequently estimate the aged admittance matrix. Specifically, we develop a GENCO-based algorithm to estimate aging-related changes in transmission line impedances. We investigate whether GENCO's learned representations can support accurate recovery of line parameters by iteratively optimizing impedance multipliers while keeping the model parameters fixed. The proposed approach reuses the existing forward model without training a dedicated inverse power-flow model or introducing an auxiliary parameter-estimation decoder.

The remainder of this paper is organized as follows. section 2 formulates the line parameter estimation problem, introduces the line aging model, and investigates GENCO's behavior under mismatched network conditions. section 3 presents the proposed impedance estimation algorithm, while section 4 evaluates its performance in recovering aged line parameters. Finally, section 5 concludes the paper.

2 Problem statement and solution approach

Previous studies have investigated conductor aging and its effects on electrical properties and reliability [4, 7]. Although maintaining accurate network parameters is important, updated information about transmission line conditions is not always available to network modeling and development teams. Detailed asset information may be restricted due to confidentiality or organizational data-sharing policies. Furthermore, direct inspection and offline measurement can be expensive, time-consuming, and may require temporary service interruptions. Although online parameter estimation techniques exist, they may depend on specialized measurement infrastructure and sufficient measurement coverage. As a result, network models may continue to rely on outdated nominal impedance values.

Transmission line parameter estimation using SCADA and PMU data has been addressed in previous studies [1]. However, we investigate an alternative approach using grid foundation models; here, the GENCO model. Instead of retraining the model, we propose a gradient-based algorithm that adjusts the nominal line impedances to better represent the aged network conditions.

Specifically, given the network topology, nominal line impedances, and available operational observations, we estimate updated resistance and reactance values by minimizing the discrepancy between observed PF quantities and those predicted by GENCO. The pretrained model parameters remain fixed, while line impedances are iteratively refined through backpropagation. The proposed approach aims to reduce reliance on direct impedance measurements and improve the accuracy of existing network models without requiring physical intervention, provided sufficient operational information is available.

A central challenge is ensuring that the estimated impedances approach their actual aged values rather than merely minimizing PF prediction error, since different impedance configurations can produce similar

network responses. To mitigate this issue, we conduct experiments across multiple aging scenarios generated using different random seeds to identify an appropriate early stopping criterion.

2.1 Line aging model

To investigate the effects of aging on GENCO's predictions, we generated an aged IEEE-118 network from its nominal configuration. Weibull distributions have been employed in previous studies to model conductor aging and failure probabilities [7, 2]. Following this approach, we assigned synthetic ages to transmission lines using a right-truncated Weibull distribution with shape parameter $k = 2.5$, scale parameter $\lambda = 40$ years, and maximum age $A_{\max} = 60$ years. The probability density function is given by:

$$f_A(a) = \frac{k\, a^{k-1} e^{-(a/\lambda)^k}}{\lambda^k \left(1 - e^{-(A_{\max}/\lambda)^k}\right)}, \quad 0 \le a \le A_{\max}. \tag{1}$$

where a denotes the transmission line age. The resulting age distribution is illustrated in Figure 1.

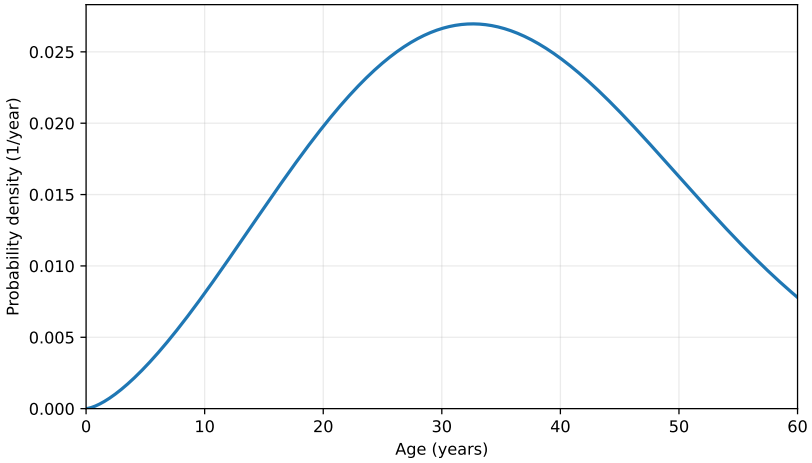


Figure 1: Right-truncated Weibull distribution used to model transmission line aging.

The assigned ages were mapped to impedance increases ranging from 0% to 50% using:

$$D_i = 0.5 \left(\frac{A_i}{60}\right)^2, \tag{2}$$

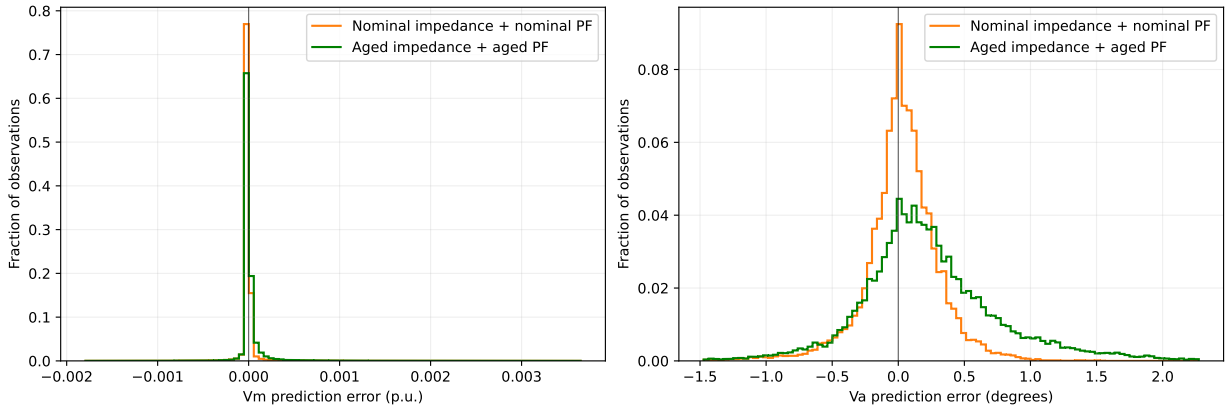
where A_i represents the age of line i. We multiplied resistance and reactance independently by the factor $(1 + D_i)$. These parameters represent a synthetic aging scenario rather than an empirically calibrated degradation model.

2.2 Solution approach using foundation models

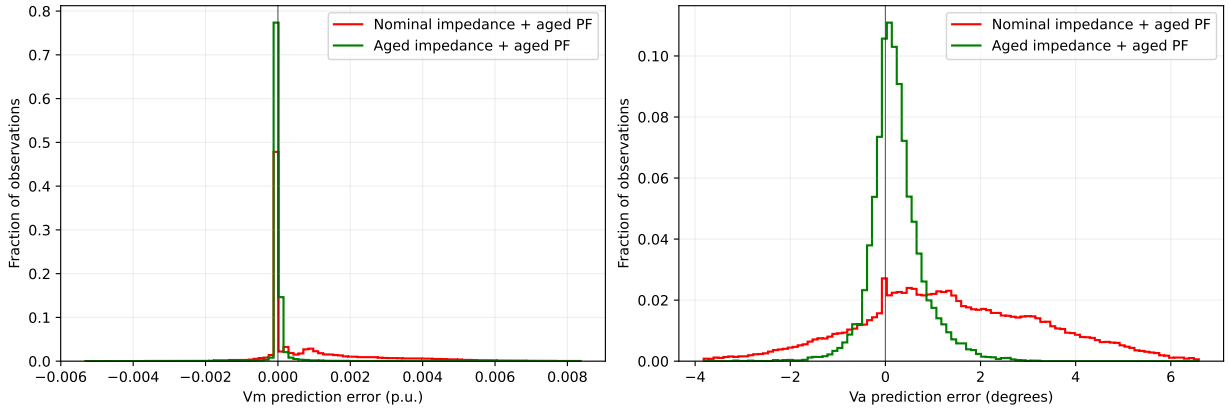
We rely on the observation that GENCO's prediction errors are approximately centered around zero when the supplied admittance matrix matches the network underlying the PF observations. We then investigate how line impedance mismatches due to aging affect GENCO's performance when predicting network states from SCADA-like PF observations generated using the aged admittance matrix.

Case 1: Matched admittance and SCADA-like PF observations

Under matched conditions, aging increases GENCO's prediction errors, particularly for voltage phase angles (V_a), whose error distribution becomes wider. However, the signed prediction errors remain approximately centered around zero, with no pronounced systematic shift, as shown in Figure 2a. Across seeds 0–4, the mean signed V_a error ranged from 0.0240° to 0.0438° under nominal matched conditions and from 0.0237° to 0.2336° under aged matched conditions. These results suggest that GENCO's prediction errors remain



(a) Matched network conditions: nominal and aged impedances.



(b) Matched and mismatched impedance conditions using aged PF observations.

Figure 2: GENCO prediction-error distributions for the IEEE-118 network under (a) aged and (b) nominal impedance conditions.

Table 1: Impedance estimation errors for IEEE-118 across seeds 0–4. Initial and final values correspond to epochs 0 and 250, respectively. RMSE values are in p.u.

	Resistance RMSE			Reactance RMSE		
Seed	Nominal	Estimated	Reduction (%)	Nominal	Estimated	Reduction (%)
0	0.0078	0.0053	32.4	0.0302	0.0181	39.8
1	0.0070	0.0051	27.4	0.0257	0.0151	40.9
2	0.0084	0.0053	36.4	0.0315	0.0184	41.6
3	0.0080	0.0054	32.2	0.0292	0.0172	41.0
4	0.0078	0.0055	29.4	0.0308	0.0171	44.5

approximately centered around zero when the supplied admittance matrix matches the network underlying the PF observations.

Case 2: Mismatched admittance and SCADA-like PF observations

In the second experiment, we investigate GENCO's response to mismatched network parameters. Specifically, we use the nominal admittance matrix to predict network states from PF observations generated using the aged admittance matrix. As shown in Figure 2b, this mismatch produces a pronounced positive shift in V_a prediction errors. Voltage magnitude (V_m) errors also increase, although the systematic shift is more evident in V_a. Across five aged networks generated using different random seeds (0–4), the mean signed V_a prediction error ranged from 1.254° to 1.866°. Notably, even the smallest mean error under mismatched conditions exceeds the largest mean error observed under matched conditions by more than a factor of five.

These observations suggest that GENCO's prediction residuals can serve as indicators of line impedance mismatch. Specifically, a systematic shift in the mean prediction error relative to a matched-condition baseline may indicate that the nominal admittance matrix no longer accurately represents the aged network. This motivates using GENCO as a surrogate PF model to detect potential impedance drift by comparing its predictions with available operational observations.

3 Impedance Estimation Algorithm

To estimate aged line impedances, we propose a gradient-based optimization method that uses GENCO as a surrogate PF model. As illustrated in Figure 3, GENCO's parameters remain frozen throughout the process, while the line resistances and reactances are treated as learnable parameters.

Rather than directly updating the impedances, independent multipliers are introduced for each eligible transmission line (bus-to-bus lines):

$$\hat{R}_l = \alpha_l R_l^0, \qquad \hat{X}_l = \beta_l X_l^0, \quad \alpha_l, \beta_l \in [1, 1.5], \tag{3}$$

where R_l^0 and X_l^0 represent the nominal line parameters. The estimated impedances are converted to admittances and supplied to GENCO alongside the aged network's operating conditions.

The optimization minimizes the mean squared error between GENCO's predicted and actual bus voltage magnitudes and angles. Gradients are backpropagated through the frozen model to update only the impedance multipliers. We use Adam with a learning rate of 3×10^{-3}, accumulating gradients over the entire dataset before each update. The multipliers are subsequently projected onto their predefined bounds.

To select the optimization hyperparameters, five independent experiments were conducted using different impedance perturbations generated with random seeds 0–4. We used the convergence behavior of resistance and reactance estimation errors, shown in Figure 4, to determine an appropriate stopping criterion and prevent overfitting.

4 Results

The proposed method is evaluated in terms of impedance estimation performance and the resulting improvement in PF predictions.

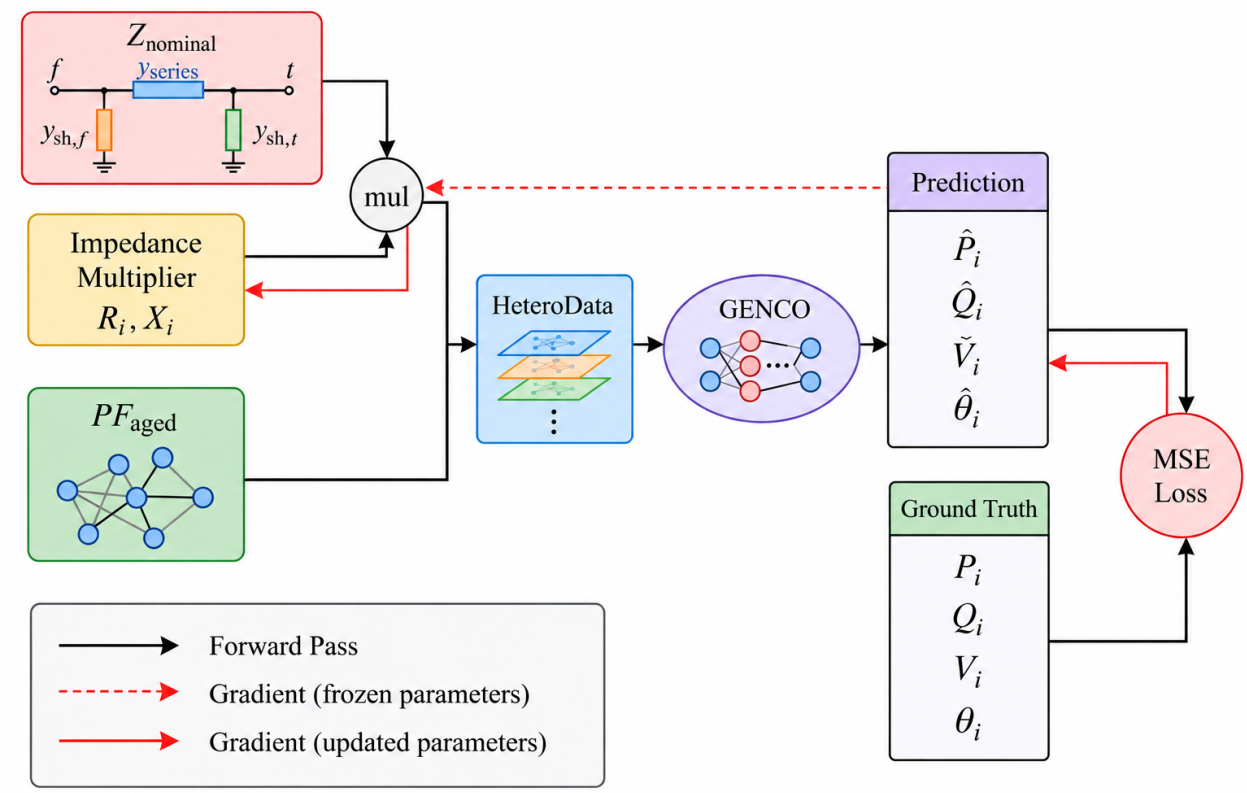
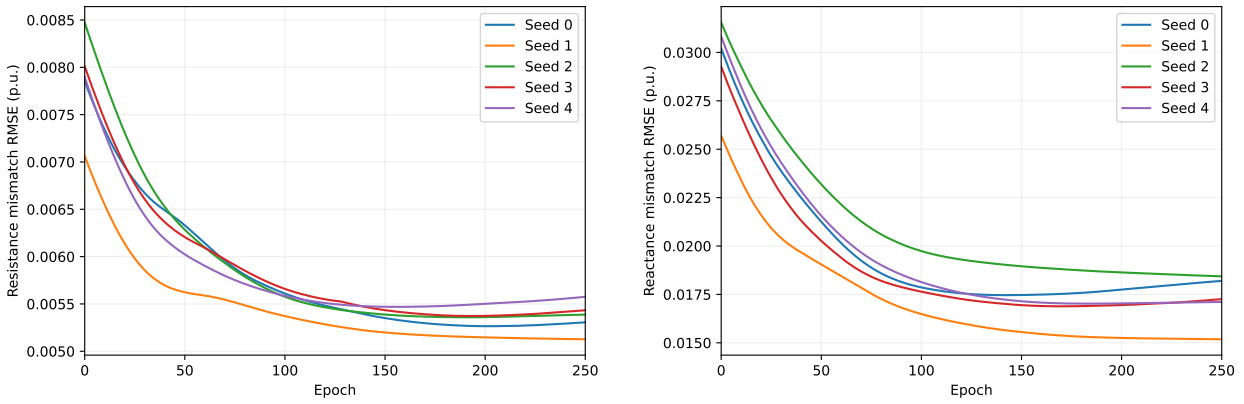


Figure 3: Proposed GENCO-based impedance estimation framework. The model parameters remain frozen while gradients are propagated to the learnable impedance multipliers.



(a) Resistance mismatch RMSE

(b) Reactance mismatch RMSE

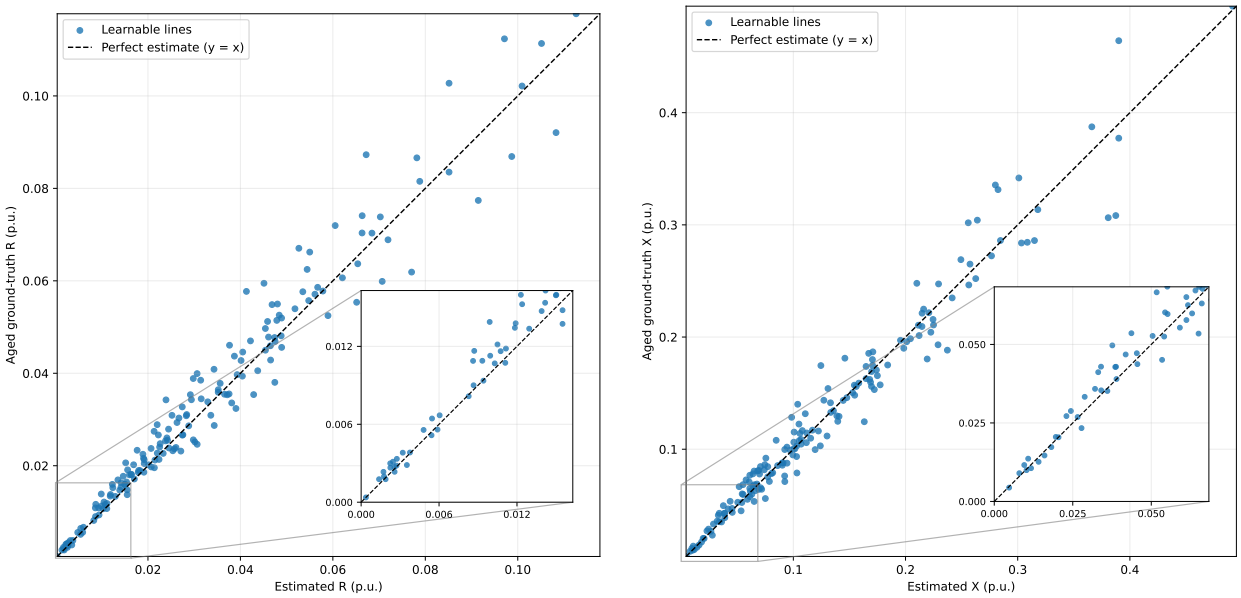
Figure 4: Evolution of impedance estimation errors during optimization across five independent experiments (seeds 0–4).

First, the optimization dynamics presented in Figure 4 demonstrate an overall reduction in impedance mismatch. For seed 0, the resistance RMSE decreases from 0.007851 to 0.005306 p.u., corresponding to a 32.42% reduction. Similarly, the reactance RMSE decreases from 0.030203 to 0.018199 p.u., representing a 39.75% improvement. Table 1 summarizes the nominal and estimated RMSE values across all five seeds. Resistance RMSE reductions range from 27.41% to 36.42%, while reactance RMSE reductions range from 39.75% to 44.47%.

Figure 5 compares the estimated resistance and reactance values against their aged counterparts for the experiment with seed 0. Most points are concentrated near the ideal $y = x$ line, indicating that the estimated parameters generally approximate the aged impedances. Nevertheless, some deviations remain, particularly for lines with larger impedance values.

Finally, Figure 6 compares GENCO's PF prediction errors under three conditions: (i) using the aged admittance matrix, (ii) using the nominal admittance matrix, and (iii) using the estimated admittance matrix. In all cases, the PF reference values correspond to the aged network.

Using the nominal admittance matrix introduces a noticeable shift and dispersion in the prediction errors, particularly for V_a. After impedance estimation, the error distributions become substantially narrower and are again concentrated around zero. Although the predictions do not fully match those obtained using the aged impedances, the results demonstrate that the proposed method can substantially mitigate the effects of unknown impedance drift.



(a) Resistance estimation

(b) Reactance estimation

Figure 5: Parity plots comparing estimated and aged line impedances for the IEEE 118-bus network (seed 4).

To evaluate the generalizability of the proposed approach, we repeated the experiment on the IEEE 57-bus network using random seed 0. The results show that the method is not limited to a specific network configuration and can be applied to other power networks, as discussed further in Appendix A.

5 Conclusion

We investigate using the grid foundation model GENCO to estimate unknown transmission line impedances affected by aging. Despite the simplicity of the proposed gradient-based approach, we showed that a pre-trained forward PF model can address the impedance drift problem of the admittance matrix. The estimated impedances agreed meaningfully with the aged line parameters and substantially improved PF predictions

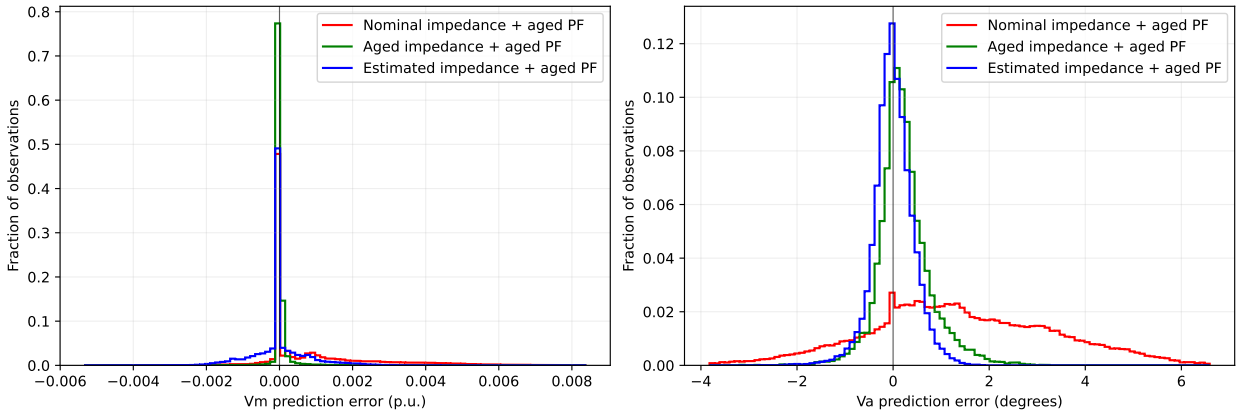


Figure 6: Distribution of power-flow prediction errors for voltage magnitude (left) and voltage angle (right). Three admittance configurations are compared: actual aged (orange), original (red), and estimated (green).

under impedance mismatch.

More importantly, these findings suggest that GENCO's capabilities extend beyond those of a conventional PF regressor. The informative gradients with respect to line impedances indicate that the model has learned useful relationships between network parameters and operating conditions. Although this does not establish full model interpretability, it provides a promising direction for investigating the physical significance of its learned representations. Future research should further explore GENCO's potential for aged PF problems, parameter identification, and gradient-based model interpretability.

References

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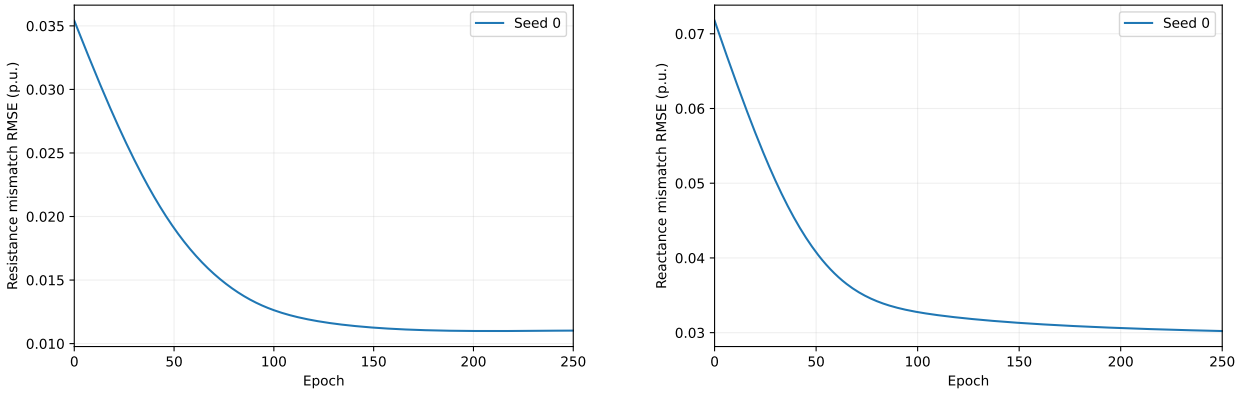
A Validation on the IEEE 57-Bus Network

To examine whether the proposed impedance estimation approach is applicable beyond the IEEE 118-bus network, we repeated the experiment on the IEEE 57-bus network using a synthetic aging scenario generated

with random seed 0. We followed the same overall procedure. The following results assess the convergence of impedance estimation, the agreement with aged line parameters, and the effect of the estimated impedances on PF prediction errors.

Impedance estimation dynamics. Figure 7 shows how resistance and reactance mismatch RMSE evolve over the optimization epochs. The resistance RMSE decreases from 0.035411 to 0.011023 p.u., corresponding to a 68.87% reduction. Likewise, the reactance RMSE decreases from 0.071755 to 0.030217 p.u., representing a 57.89% reduction.

The IEEE 57-bus network has higher average line impedances than the IEEE 118-bus network, exhibiting impedance characteristics closer to those typically associated with distribution networks. Consequently, the same percentage increase in impedance due to ageing results in larger absolute deviations in line parameters. The greater improvement in impedance estimation observed for the IEEE 57-bus network suggests that (i) the proposed algorithm performs effectively when impedance mismatches are substantial and (ii) the method is applicable across power networks with different impedance characteristics.



(a) Resistance mismatch RMSE

(b) Reactance mismatch RMSE

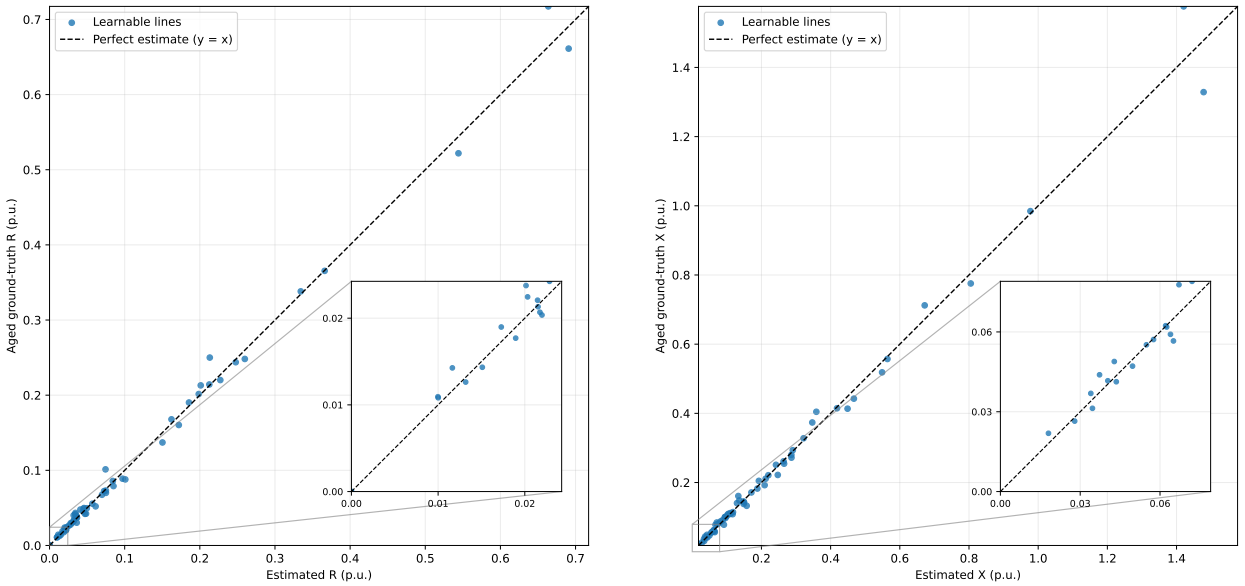
Figure 7: Evolution of the line impedance estimation errors for the IEEE 57-bus network (seed 0).

Agreement with the aged line impedances. Figure 8 compares the estimated resistance and reactance values with the corresponding aged values. Most points lie close to the ideal $y = x$ line, suggesting that the recovered parameters agree well with the aged network. Comparing the parity plots of the IEEE 57-bus and IEEE 118-bus networks reveals a more pronounced improvement in estimation accuracy for the IEEE 57-bus case, which may be attributed to the aforementioned differences in nominal line impedances between the two networks.

PF prediction errors under matched and mismatched conditions. We first compare GENCO's prediction error distributions for the two matched conditions: nominal admittances with nominal PF observations, and aged admittances with aged PF observations. As shown in Figure 9, both conditions exhibit small systematic offsets, particularly in the V_a errors. These offsets are present even when the admittance matrix matches the network underlying the observations. They therefore reflect a baseline imperfection in GENCO's predictions rather than an error introduced by the proposed impedance estimation procedure. Accordingly, the estimation objective is not necessarily to make all prediction errors exactly zero, but to mitigate the additional errors associated with using mismatched impedances.

When aged PF observations are supplied alongside nominal admittances, the resulting mismatch produces substantially broader prediction error distributions, as illustrated in Figure 10. In contrast, supplying the aged admittances yields much more concentrated distributions, although the small baseline offsets noted above remain.

Finally, Figure 11 compares three cases using the same aged PF observations: nominal, aged, and estimated admittances. The estimated admittances substantially narrow the voltage magnitude and phase-



(a) Resistance (b) Reactance

Figure 8: Parity plots of estimated versus aged line impedances for the IEEE 57-bus network (seed 0). The insets magnify the low-impedance region.

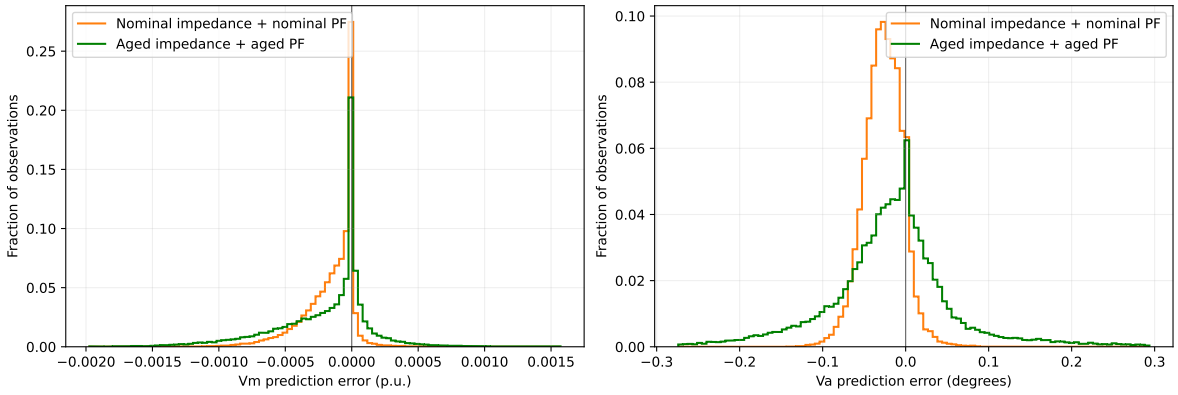


Figure 9: GENCO prediction-error distributions for nominal and aged matched conditions on the IEEE 57-bus network: voltage magnitude (V_m, left) and phase angle (V_a, right).

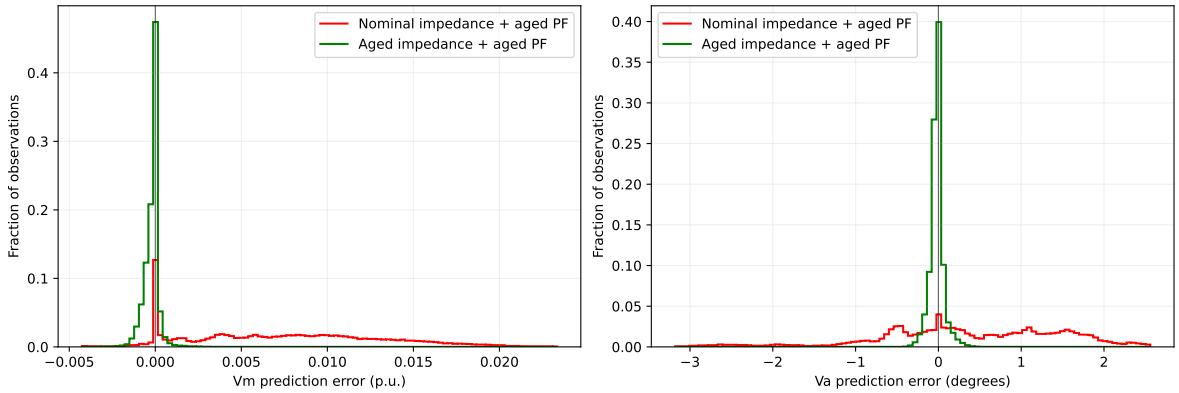


Figure 10: GENCO prediction errors for the IEEE 57-bus network under mismatched (nominal admittances with aged PF observations) and matched (aged admittances with aged PF observations) conditions.

angle prediction error distributions relative to the nominal admittance case, bringing them closer to the aged-admittance reference. The V_m prediction error decreases by 93.04%, while the V_a prediction error decreases by 92.90%, according to RMSE computed over all 57 buses. This improvement is achieved despite the non-zero baseline prediction errors observed under matched conditions. Together with the IEEE 118-bus results, the IEEE 57-bus experiment provides additional evidence that the approach is not restricted to a single network configuration.

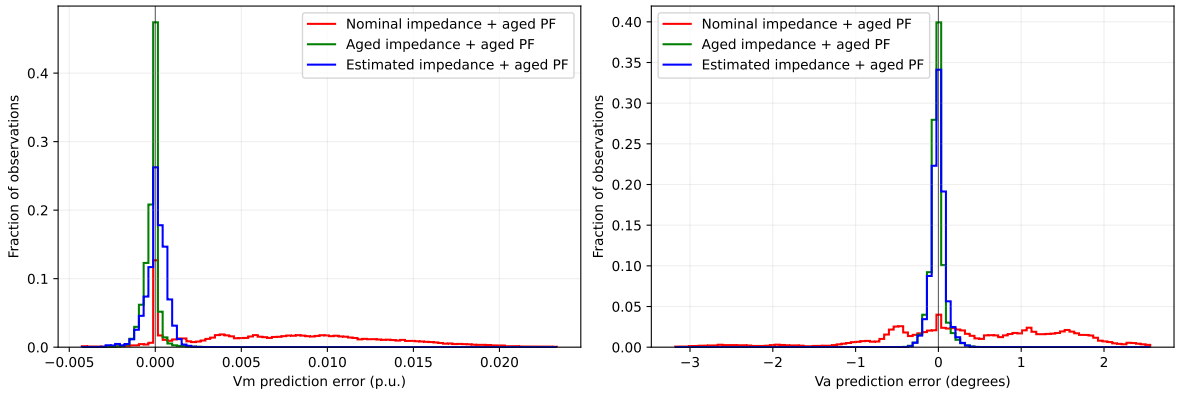


Figure 11: V_m and V_a prediction error distributions for the IEEE 57-bus network.

B GENCO Error Bias on IEEE 57-Bus

We observed a shift in the signed-error histogram in Figure 9. Because we used the same configuration file to generate the 118- and 57-bus results, we further investigated the root cause of this inconsistency.

Three sources of perturbation in the original data-generation process affect GENCO training. Based on [6], they used: a) topology perturbation to add $N - k$ topology variations where $k >= 1$ of the lines or generators are removed randomly, b) generation cost perturbation so that the dataset contains various levels of power generation at different nodes, and c) line admittance perturbation, so that the model is trained in a robust way to its reference network and to figure out physics constraints rather than a specific network. We deliberately disabled these perturbations to reduce noise in the GENCO prediction error caused by non-aging factors.

To pinpoint each perturbation's effect on the skewness of the signed error around zero, we run a set of experiments. To verify that our data generation using datakit GENCO is correct, we fetch the power

flow data created by the GENCO development team and separate the test partition to plot the model's performance. The original data is named $pf_small_case57_ieee$ and can be found at this address.

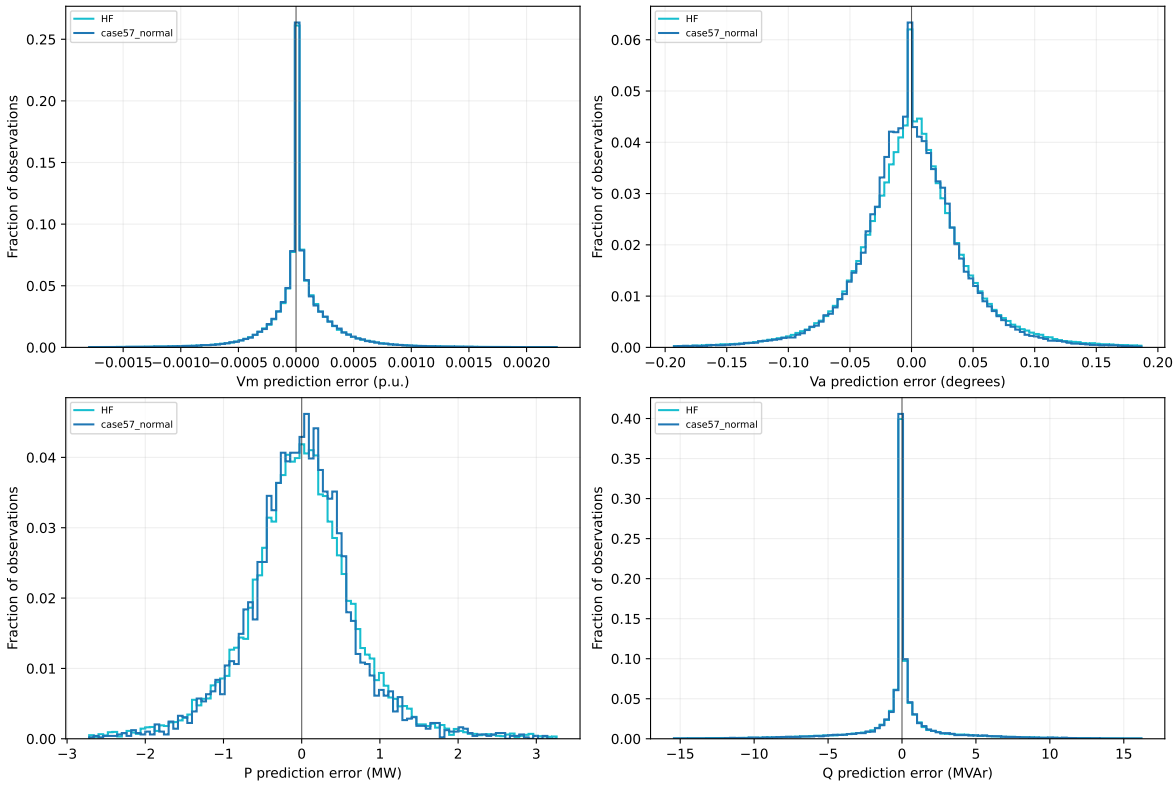


Figure 12: Prediction error distribution of the GENCO test set VS our generated dataset.

To measure how much the error is shifted from the center point, $x = 0$, we used a zero-centered skewness, as follows:

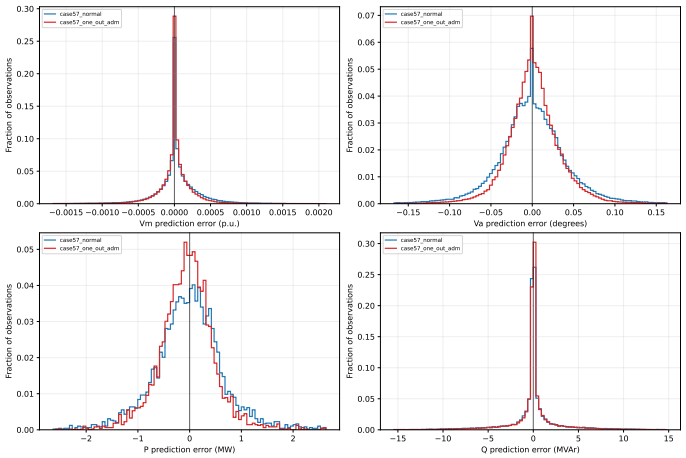
$$\text{Skewness}_0 = \frac{\frac{1}{N}\sum_{i=1}^{N} x_i^3}{\left(\frac{1}{N}\sum_{i=1}^{N} x_i^2\right)^{3/2}} \tag{4}$$

where N is the number of samples in the dataset. The general form of skewness is defined around the mean of the data. But in neural network model with physics informed property, we expect the error has no bias and be around the $x = 0$. Any shift of error around zero is not acceptable and should be accounted for in model design or during the training process.

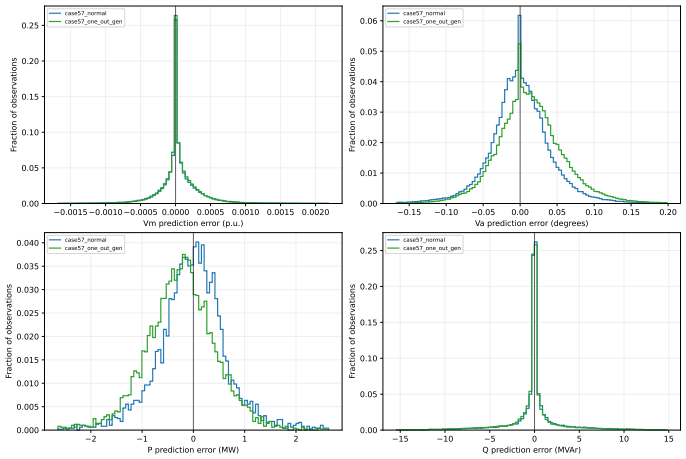
Figure 12 shows that the prediction error of the GENCO model using our generated dataset and the GENCO test set has no significantly different skewness. The skewness for the GENCO test set with 21694 samples reads as 12.90, 0.26, 1.35, and -0.29, for V_m, V_a, P, and Q predictions, respectively. These values for our dataset, generated with 4930 samples, read as -0.14, 2.16, 0.42, and -0.28. This experiment proves that GENCO has no error bias on its original dataset and our generated data.

In the next set of experiments, we only remove one type of perturbation each time and generate data. This makes the underlying distribution of the generated dataset one step closer to the data we used to estimate the line admittance. The one_out prefix in the names of the datasets represents the type of perturbation removed from data generation.

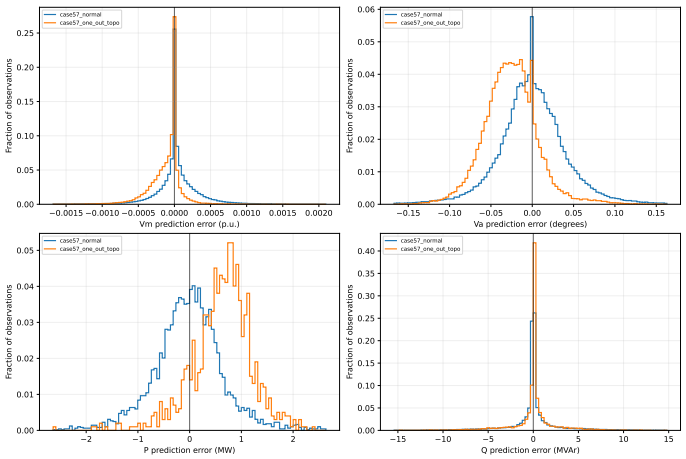
Plotting these error distributions against each other in separate plots in Figure 13, we can observe that removing generation cost and topology perturbations adds some bias to the error of the model, which was not aligned with our hypothesis that the model error has no bias. However, removing admittance perturbation does not add any bias to the prediction error.



(a) Admittance perturbation



(b) Generation cost perturbation



(c) Topology perturbation

Figure 13: Comparison of GENCO prediction errors under exclusion of different perturbation types in the IEEE 57-bus network.

For the last experiment regarding GENCO prediction error evaluation, we disabled all three kinds of perturbation and generated an all_out dataset, similar to what was used in our original admittance estimation experiment.

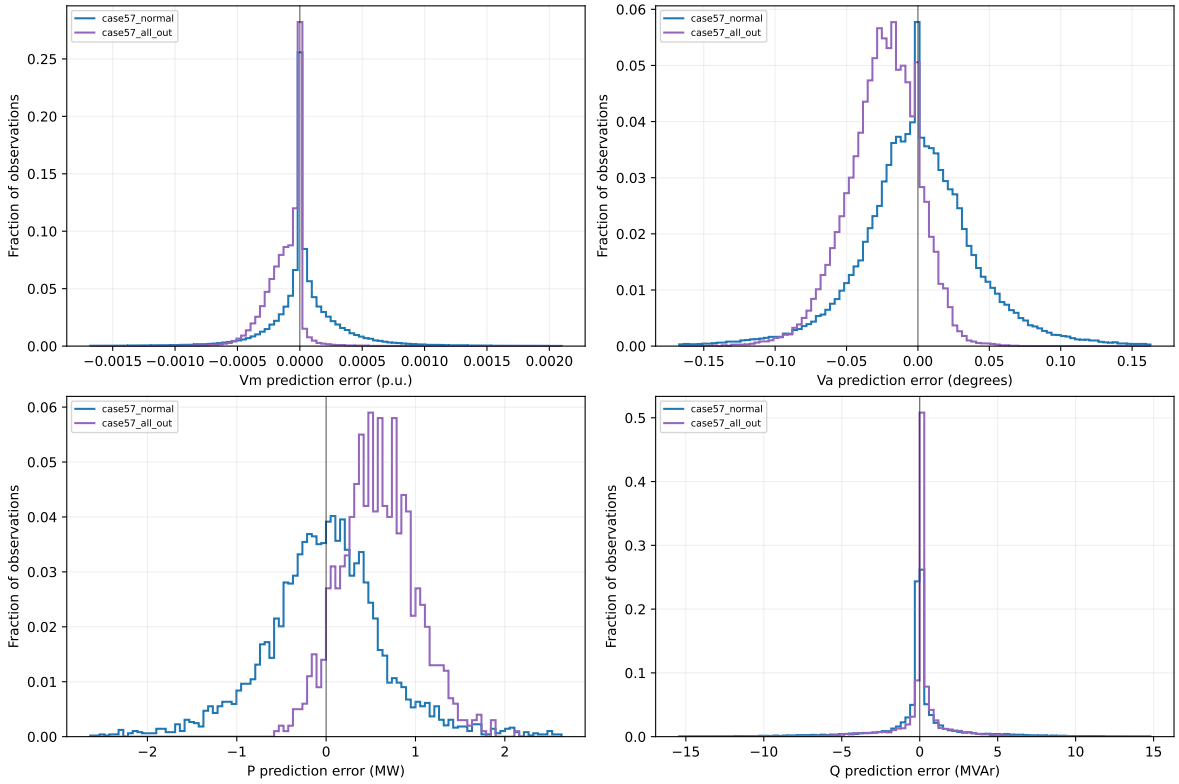


Figure 14: Prediction error distribution of the normal dataset vs. the all-perturbation-disabled dataset.

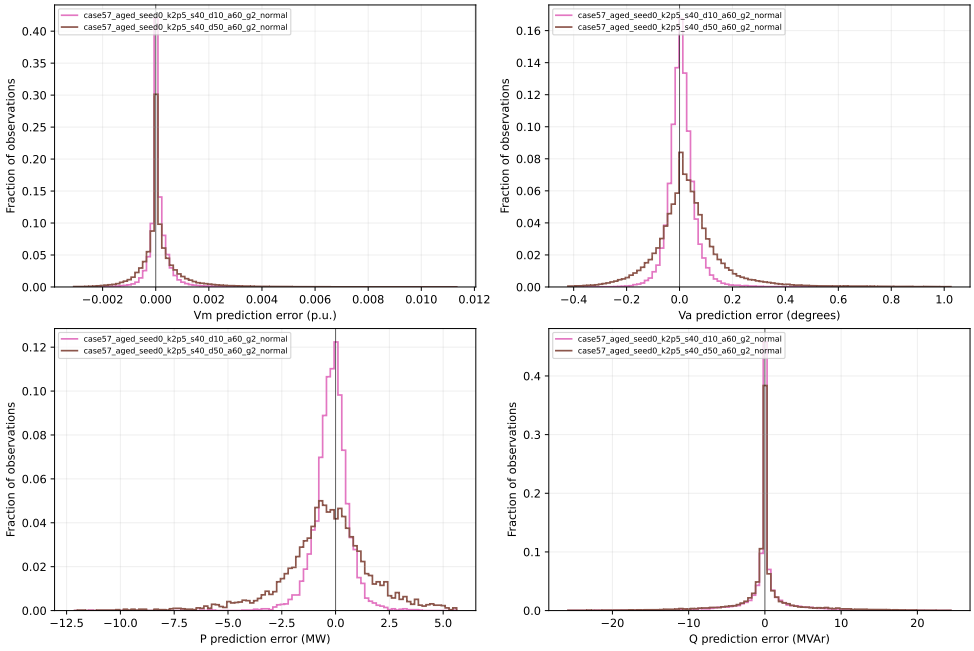
As illustrated in Figure 14, there is a clear prediction error in the out-of-distribution data that was generated without any perturbation. The skewness around zero for this dataset is measured as -1.81, -1.57, 1.40, and -0.81 for V_m, V_a, P, and Q predictions, respectively. Table TODO summarizes the skewness around zero for all the above-mentioned experiments.

C Sensitivity of proposed method on prediction error bias

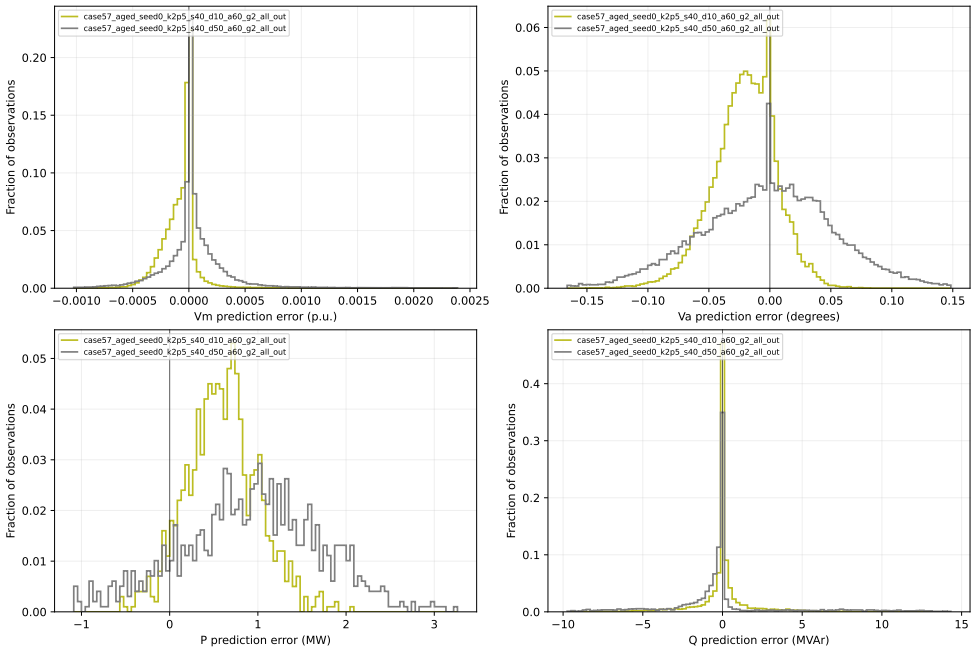
In Appendix B, we showed that removing some types of perturbation caused the prediction error to have biases, which should be taken into consideration and addressed in the future development of GENCO. This breach in the existing condition of the GENCO model drives us to further analyze our algorithm on an unbiased dataset with all kinds of perturbations. More specifically, we activate topology, generation cost, and admittance perturbations in our data generation process to obtain an unbiased prediction error model. To further scrutinize our model training process in the presence of all types of perturbations, we used two types of aging: weak and normal aging conditions. In weak aging, we sampled the age of the lines from Equation 1 and softened the aging to add the impedance up to 10%, while in normal aging, this number is set to 50%. To make the comparison complete, we also generate the same aged datasets twice, with all active and all inactive perturbations.

As expected, Figure 15 shows that removing perturbation in data generation adds bias to the prediction error, and Figure 15b shows this clearly, while we can not observe noticeable skewness around zero error in Figure 15a. The takeaway is that, in both cases, skewness around zero for P and Q is higher in severe cases.

During impedance estimation, the GENCO prediction error decreases continuously. In Figure 16a, we can see that the normal aging cases, with up to a 50% increase in resistance or reactance, start with higher

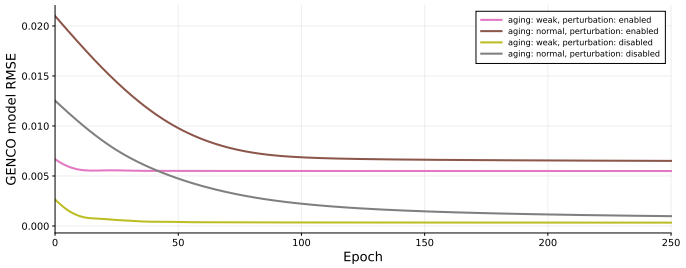


(a) All perturbations enabled.

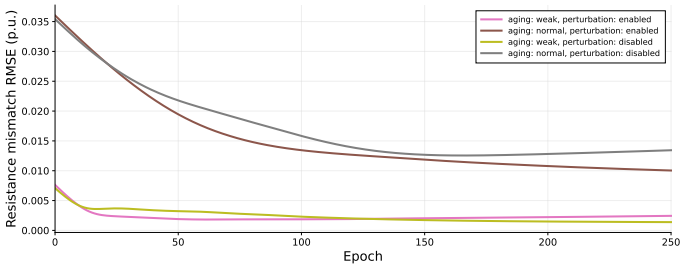


(b) All perturbations disabled.

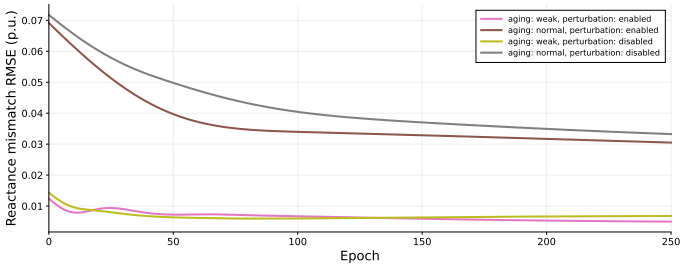
Figure 15: Comparison of GENCO prediction error distributions under weak and normal aging conditions in the IEEE 57-bus network. The distributions of the prediction errors for V_m, V_a, P, and Q are compared under (a) enabled and (b) disabled perturbations.



(a) GENCO model RMSE.



(b) Resistance estimation RMSE.



(c) Reactance estimation RMSE.

Figure 16: Evolution of GENCO model RMSE and impedance estimation errors during optimization in the IEEE 57-bus network.

RMSE values. Also, in Figure 16b and Figure 16c, the normal-aging curves start at higher RMSE values, as expected. Table 2 shows the relative improvement in resistance and reactance estimation.

Table 2: Relative improvement in resistance and reactance estimation under different aging and perturbation conditions in the IEEE 57-bus network.

Ageing	**Perturbations**	**Relative RMSE improvement for R**	**Relative RMSE improvement for X**
Weak	Enabled	68.1%	60.0%
Normal	Enabled	72.1%	55.9%
Weak	Disabled	80.4%	52.7%
Normal	Disabled	62.1%	53.7%

In conclusion, the experiments in this section show that the algorithm works well for both small and severe aging effects, and it works well even when the GENCO model prediction error violates the unbiased conditions.